

Solution

Q.1

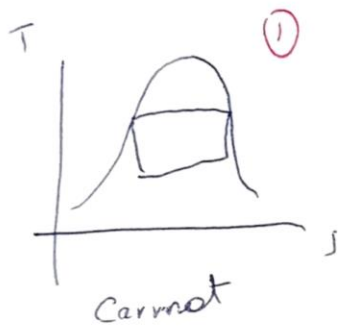
Primary Energy: Available in nature before any conversion (0.5)
Example: Crude oil, sunlight (0.5)

Final Energy: Energy delivered to consumer in usable form (0.5)

Example: Electricity, petrol (0.5)

Q.2 $CF = \frac{1.1 \times 10^7}{1500 \times 10^3 \times 8760} = 0.0837 \approx 8.4\%$ (1) ECF

Q.3



Q.4 $\eta = 1 - \frac{T_c}{T_H} \Rightarrow 0.9 = 1 - \frac{273+27}{T_H} \Rightarrow T_H = \frac{300}{0.1} = 3000^\circ K$ (1)
 $= 2727^\circ C$
(0.5) (0.5)

Q.5 $\eta_{cc} = 1 - (1 - \eta_1)(1 - \eta_2)$
 $= \eta_1 + \eta_2 - \eta_1 \eta_2$ (2)

Q.6 Natural gas (1)

Q.7 $\eta = 1 - \frac{T_c + 273}{T_H + 273}$
(1)

Coal	Gas
Rankine cycle Coal as fuel Steam cycle High capital Cheaper to operate Base load More pollutants	Brayton cycle Natural gas as fuel Direct combustion Low capital More expensive to operate Peak load Less pollutants

② for 2 correct

① for every incorrect comparison

Q.9 $LCOC = I_0 + \sum_{t=1}^n \frac{A_t}{(1+i)^t}$

Envy produced

$$= (500 \times 10^6 \times 0.500 \times 10^{-3}) + \sum_{t=1}^{20} \left(\frac{40}{10^3} \times 500 \times 10^6 + 25 \times 10^{-6} \times 500 \times 10^6 \times 8760 \times 0.85 \right) \frac{1}{(1.1)^t}$$

$$20 \times 0.85 \times 500 \times 10^6 \times 8760$$

$$= 1.25 \times 10^9 + \sum_{t=1}^{20} \left[\frac{20 \times 10^6 + 93.075 \times 10^6}{(1.1)^t} \right]$$

$$74.46 \times 10^{12}$$

$$= \frac{1.25 \times 10^9 + \sum_{t=1}^{20} \frac{113.075 \times 10^6}{(1.1)^t}}{74.46 \times 10^{12}}$$

$$74.46 \times 10^{12}$$

$$= \frac{1.25 \times 10^9 + 0.963 \times 10^9}{74.46 \times 10^{12}} \approx 3 \times 10^{-5} \$/Wh$$

$$\approx 3 \$/kWh$$

$$\approx 30 \$/MWh$$